LED HOUSE — ELECTRICAL SHOP

Functional Requirements

Requirements Traceability Matrix (RTM)

Document Type	Functional Requirements Catalog / RTM
Project	LED House — Web-based E-Commerce System
Version	1.0
Total Functional Req.	43 (FR_01–FR_43)
Total Non-Functional Req.	9 (NFR_01–NFR_09)
Total External Req.	2 (ER_01–ER_02)

1. Introduction

This document is the Functional Requirements Catalog and Requirements Traceability Matrix (RTM) for the LED House e-commerce system. It provides a structured technical inventory of all requirements, their priority, assigned actor, dependencies, and a cross-reference link to the companion Legal-Functional Requirement Specification (LFRS) document where applicable.

Each requirement is assigned a Priority Level based on system criticality:

- P1 – Critical: Required for the system to function. Failure = system unusable or legally non-compliant.
- P2 – High: Required for full feature delivery. Omission significantly degrades the product.
- P3 – Medium: Enhances usability and business value. May be deferred to a later release.

2. Functional Requirements (FR)

The following tables catalogue all 43 functional requirements, grouped by module. The 'LFRS Link' column references the corresponding entry in the Legal-Functional Requirement Specification document where a legal obligation applies.

2.1 Product Management

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_01	**Create Product**	Admin enters product name, description, category, purchase price, selling price, and stock quantity to create a new product record.	*Admin*	**P1 – Critical**	—	*Core operation*
FR_02	**Validate Product Data**	System enforces: name not empty, prices > 0, stock ≥ 0	*System*	**P1 – Critical**	—	*Input guard*

ID	Title	Description	Actor	Priority	LFRS Link	Notes
		before saving a product.				
FR_03	**Edit Product**	Admin can modify any field of an existing product record.	*Admin*	**P1 – Critical**	—	—
FR_04	**Delete Product**	Admin can permanently remove a product from the catalogue.	*Admin*	**P2 – High**	—	*Soft-delete preferred*
FR_05	**View Product List**	All users can view a paginated list of products with name, price, and availability status.	*User / Admin*	**P1 – Critical**	*LFRS-10*	—
FR_06	**Product Details View**	Users can view the full detail page of a specific product.	*User*	**P1 – Critical**	—	—
FR_07	**Product Search**	Users can search for products using a keyword input field.	*User*	**P2 – High**	—	—
FR_08	**Product Filtering**	Users can filter the product list by category and price range.	*User*	**P2 – High**	—	—

2.2 Inventory Management

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_09	**Track Stock Quantity**	System maintains a real-time stock count per product.	*System*	**P1 – Critical**	*LFRS-10*	—
FR_10	**Automatic Stock Update**	System automatically decrements stock count after a successful order is placed.	*System*	**P1 – Critical**	*LFRS-10*	—
FR_11	**Prevent Out-of-Stock Orders**	System blocks order placement if product stock count is zero. User is shown an availability notice.	*System*	**P1 – Critical**	*LFRS-10*	*Legal + Technical*
FR_12	**Low Stock Notification**	System alerts the admin when any product falls below a configurable minimum threshold.	*System*	**P2 – High**	—	—

2.3 Customer Management

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_13	**User Registration**	Users create an account by providing name, email or phone, and password.	*User*	**P1 – Critical**	*LFRS-01*	*Consent required*
FR_14	**User Authentication**	Registered users log in using their email/phone and password.	*User*	**P1 – Critical**	*LFRS-01, LFRS-14*	—
FR_15	**Password Security**	Passwords are stored using a cryptographic hash algorithm (not plaintext).	*System*	**P1 – Critical**	*LFRS-02*	*MANDATORY – Law 124/2024*
FR_16	**View Profile**	Authenticated users can view and edit their own account information.	*User*	**P2 – High**	*LFRS-05*	*Right of access*

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_17	**Order History**	Users can view a chronological list of their past orders with status.	*User*	P2 – High	—	—

2.4 Shopping Cart & Checkout

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_18	**Add to Cart**	Authenticated users can add products to a session-based shopping cart.	*User*	P1 – Critical	—	—
FR_19	**Modify Cart**	Users can update item quantities or remove items from the cart before checkout.	*User*	P1 – Critical	—	—
FR_20	**Cart Summary**	Cart page displays all selected products, individual quantities, unit prices, and total price.	*User*	P1 – Critical	*LFRS-09*	*Legal pricing display*
FR_21	**Checkout Process**	User proceeds from cart to checkout, entering delivery and payment details.	*User*	P1 – Critical	*LFRS-06, LFRS-09*	—
FR_22	**Order Confirmation**	User must confirm the final order summary before the transaction is finalised.	*User*	P1 – Critical	*LFRS-08*	*Pre-contractual info*

2.5 Order Management

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_23	**Create Order**	System generates a persistent order record after the checkout is completed.	*System*	P1 – Critical	*LFRS-08*	—
FR_24	**Order Status Tracking**	Each order has a status: Pending, Confirmed, or Delivered. Status is visible to the customer.	*System / Admin*	P1 – Critical	—	—
FR_25	**Manual Order Confirmation**	Admin can manually change an order status from Pending to Confirmed.	*Admin*	P1 – Critical	—	—
FR_26	**View All Orders (Admin)**	Admin can view a list of all customer orders with their current statuses.	*Admin*	P1 – Critical	*LFRS-11*	—

2.6 Payment Processing

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_27	**Payment Method Selection**	During checkout, users select one of: PayPal, Card via 2Checkout (Verifone), or Cash on Delivery.	*User*	P1 – Critical	*LFRS-06, LFRS-07*	—
FR_28	**Online**	For PayPal and 2Checkout	*System*	P1 –	*LFRS-*	—

ID	Title	Description	Actor	Priority	LFRS Link	Notes
	Payment Redirection	selections, the system redirects users to the external payment gateway.		**Critical**	*06*	
FR_29	**Payment Confirmation**	System records successful payment events and updates the corresponding order.	*System*	**P1 – Critical**	*LFRS-08, LFRS-11*	—
FR_30	**COD Support**	Orders can be placed and confirmed without an online payment. Payment is handled at delivery.	*User*	**P1 – Critical**	*LFRS-07*	*Albanian market need*

2.7 Service / Repair Management

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_31	**Submit Service Request**	Customers can submit a repair or service request through the system interface.	*User*	**P2 – High**	*LFRS-12*	—
FR_32	**Service Details Storage**	System stores the product description, issue description, and customer contact for each service request.	*System*	**P2 – High**	*LFRS-12*	*Personal data – law applies*
FR_33	**Service Status Tracking**	Each service request has a status: Pending, In Progress, or Ready. Visible to the customer.	*System / Admin*	**P2 – High**	—	—
FR_34	**Update Service Status**	Admin can change the status of any service request.	*Admin*	**P2 – High**	—	—
FR_35	**Estimated Completion Time**	Admin can set and update an expected completion date for a service request.	*Admin*	**P3 – Medium**	—	—

2.8 Notification System

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_36	**Order Notifications**	System notifies the customer when their order is confirmed and when it is delivered.	*System*	**P2 – High**	*LFRS-04*	*Consent required*
FR_37	**Service Notifications**	System notifies the customer when their repair/service is completed.	*System*	**P2 – High**	*LFRS-04*	—
FR_38	**Notification Methods**	Notifications are delivered via email and/or phone (SMS or manual call).	*System / Employee*	**P2 – High**	*LFRS-04*	*Consent required – Law 124/2024*

2.9 Sales Analytics

ID	Title	Description	Actor	Priority	LFRS Link	Notes

ID	Title	Description	Actor	Priority	LFRS Link	Notes
FR_39	**Sales Tracking**	System tracks the number of units sold per product and the period in which they were sold.	*System*	P2 – High	*LFRS-11*	—
FR_40	**Revenue Calculation**	System calculates total sales revenue over a user-selected period.	*Admin*	P2 – High	*LFRS-11*	—
FR_41	**Profit Calculation**	System calculates profit per product and total profit based on purchase price and selling price.	*Admin*	P2 – High	*LFRS-11*	*Requires purchase price stored*
FR_42	**Product Performance Analysis**	System identifies best-selling and low-performing products based on unit sales data.	*Admin*	P3 – Medium	—	—
FR_43	**Reporting**	Admin can generate and view basic sales reports covering orders, revenue, and product performance.	*Admin*	P3 – Medium	*LFRS-11*	—

3. Non-Functional Requirements (NFR)

Non-functional requirements define the quality attributes and operational constraints of the system.

ID	Title	Description	Category	Priority	LFRS Link
NFR_01	**Data Encryption**	Sensitive user data (passwords, payment info) shall be encrypted at rest and in transit.	*Security*	P1 – Critical	*LFRS-02, LFRS-14*
NFR_02	**Secure Authentication**	System shall use secure session management and prevent common authentication attacks.	*Security*	P1 – Critical	*LFRS-14*
NFR_03	**Response Time**	95% of user-facing page requests shall complete within 3 seconds under normal load.	*Performance*	P2 – High	—
NFR_04	**Usability**	The interface shall be operable by non-technical users without prior training.	*Usability*	P2 – High	—
NFR_05	**Responsive Design**	System shall be accessible and fully functional on mobile, tablet, and desktop devices.	*Accessibility*	P2 – High	—
NFR_06	**Data Storage**	System shall persistently store product, customer, order, and service data in a relational database.	*Storage*	P1 – Critical	*LFRS-03, LFRS-12*
NFR_07	**Data Backup**	System shall perform automated daily backups of all stored data.	*Storage*	P1 – Critical	*LFRS-13*
NFR_08	**System Availability**	System shall maintain at least 99% uptime during business operating hours.	*Availability*	P1 – Critical	—
NFR_09	**Admin Maintainability**	Product catalogue and content updates shall be performable by the admin without developer involvement.	*Maintainability*	P2 – High	—

4. External Requirements (ER)

External requirements define third-party integrations and legal compliance obligations outside the core system.

ID	Title	Description	Category	Priority	LFRS Link
ER_01	**PayPal & 2Checkout Integration**	System shall integrate with PayPal and 2Checkout (Verifone) as external payment providers.	*Payment*	**P1 – Critical**	*LFRS-06*
ER_02	**Legal Compliance — Albania**	System shall comply with Albanian consumer protection law, data protection law (Law 124/2024), and fiscal/tax regulations.	*Legal*	**P1 – Critical**	*LFRS-01 to LFRS-14*

5. Requirements Summary by Priority

The table below provides a consolidated count of requirements by priority level across all categories.

Category	P1 – Critical	P2 – High	P3 – Medium	Total
Functional (FR)	26	12	5	**43**
Non-Functional (NFR)	4	5	0	**9**
External (ER)	2	0	0	**2**
TOTAL	32	17	5	**54**

6. Notes on Traceability

Requirements marked with an LFRS Link must be read in conjunction with the LED House Legal-Functional Requirement Specification (LFRS) document. The LFRS link indicates that the requirement carries a legal obligation under Albanian law and that compliance validation is required before marking the requirement as complete during development.

Requirements without an LFRS link are driven by business logic, usability, or technical correctness — but have no direct legal compliance obligation attached.